

**Metric Justification and Clinical Relevance**

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## Summary of the Notebook

My notebook shows the baseline machine learning model and confusion matrix diagrams for the dummy model, logistic regression and the decision tree. The logistic regression and decision tree had accuracies higher than the dummy model, however, none of the models were able to get a good reading on ESI 1. Since most of the data is commonly attributed to ESI 3, it's only natural that the model would be more skewed to produce such a result.

The data shows that there are 55,121 patients. From those patients, only 11,025 were used in the test set. Here are the results:

| <i>Model</i>        | <i>Accuracy</i> | <i>Macro F1</i> | <i>Recall on ESI 1</i> |
|---------------------|-----------------|-----------------|------------------------|
| Dummy               | 0.371           | 0.199           | 0                      |
| Logistic Regression | 0.647           | 0.446           | 0                      |
| Decision Tree       | 0.556           | 0.225           | 0                      |

## The Recall Metric

As a result of the following results, the primary metric chosen would be recall on ESI 1, the most critical cases. In triage, a false negative and a false positive carry vastly different costs. If the model flags a stable patient as more urgent than they are, a clinician can waste time by reassessing them. Although annoying, and can cause clinician fatigue, it's always recoverable. On the other hand, if the model fails to flag a truly critical patient, like someone having organ failure, sepsis or heavy bleeding, and that patient waits in the queue behind less sick people their condition is most likely to deteriorate at an alarming rate. As such, recall directly measures based on this question, "Of the patients who were actually critical, how many did the model catch?", which is exactly the quantity that matters when the cost of missing someone is measured in a matter of minutes of their untreated condition, rather than clinician inconvenience.

As mentioned before, the dataset is skewed to have most of the patients be within ESI 3, hence why all models, besides the dummy, have such high accuracies. In this case, ESI 1 has 16 patients, which is less than 1 percent of this dataset. This shows that the model fully missed all 16 chances of classifying an ESI level 1 correctly and still had an accuracy greater than a coinflip. In the same vein, precision effectively tells you how much you can trust a positive flag, but says nothing about how many true emergencies were missed and never flagged at all. While Macro F1 averages both precision and recall to give a better view of the model's actual accuracy, the regular clinician doesn't care about a blended score. They more so care about if the system will tell them when someone is dying from when they're not.

As such, recall on the highest acuity class should be the only figure that answers that for them. Hence why it should be the metric that this overall model is judged against instead of overall accuracy and F1.

As seen in the table, all models, including the dummy, scored 0 for recall on ESI 1. Never did any model predict the most critical case in the entire set. The model doesn't just get these wrong, but it seems to disregard the category and focus more on the data that's more prevalent amongst the set. What confirms this is the decision tree's matrix output. It never predicted that any case was at ESI 1. For the model to assign a level 2 or 3 to a patient that needs attention immediately it could cost many lives. Even in a low-resource ED, like in the Caribbean, the difference between a patient being seen now versus 10 minutes later can be either life or death.

Link to Video Explainer: [🔥 AI\\_Triage\\_\\_Catch\\_Rate\\_Crisis.mp4](#)